Taizhen Cheung

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Technical Skills

Languages: C++, Python, JavaScript (ES6+), TypeScript, HTML5, CSS3, SQL
Frameworks & Libraries: React, Next.js, Node.js, Express.js, Material UI, Vite
Developer Tools: Git, GitHub, Linux/Unix, REST APIs, PostgreSQL, Supabase, Vercel, Agile Development
Concepts: Data Structures, Algorithms, Object-Oriented Programming, System Design, REST APIs, Data Analysis

Education

Hunter College, City University of New York — Expected May 2027
Bachelor of Arts – Double Major in Computer Science and Mathematics

- **Relevant Coursework:** Data Structures, Algorithms, Software Analysis & Design, Numerical Methods, Linear Algebra, Multivariable Calculus, Discrete Mathematics

Experience

AI Trainer – Software Engineering & Coding Domain — June 2025 – Present
Handshake AI – Remote

- Evaluated and scored responses from frontier large language models on multi-turn software-engineering and coding tasks, assessing correctness, helpfulness, and communication quality against detailed rubrics.
- Authored written rationales for each score and conducted comparative preference evaluations across multiple candidate responses to identical coding prompts.
- Maintained high inter-rater agreement through a rigorous calibration screening, generating high-quality human-feedback data to improve the reasoning and response quality of AI coding assistants.

Research Intern – LLM Bias and Benchmarking — June 2025 – Sep 2025
TIER Lab, Hunter College – New York, NY

- Conducted comparative analysis of 1,000+ prompts across Gemini 2.0 Flash, DeepSeek R1, and GPT-4o to evaluate reasoning consistency, response safety, and model bias.
- Designed bias audit protocols for LLM-integrated robotics systems, identifying critical failure modes in Human-Robot Interaction (HRI) safety and fairness.
- Engineered reproducible benchmark datasets using Python for robotic perception and dialogue evaluation across multiple model architectures.
- Quantified latency and bias trade-offs between high-speed and reasoning-heavy models, delivering data-driven mitigation strategies to research stakeholders.

Capital One First-Gen Focus Fellow — Oct 2024 – Dec 2024
Capital One

- Selected for competitive 8-week technical development program; collaborated with software engineers on system design principles and software development lifecycle practices.

Projects

Onward – AI Emotional-Companion Web App — May 2026 – Present
Next.js 15, TypeScript, Supabase/Postgres/pgvector, Cerebras GPT-OSS 120B, Vercel

- Built and deployed a full-stack Next.js 15 (App Router) + TypeScript application to Vercel, featuring server-side SSE streaming of narrative content and a client-side story-player state machine.
- Designed and built a custom six-lane retrieval-augmented-generation pipeline (“FacetsRAG”) combining vector search, reciprocal-rank fusion, and LLM reranking, reaching ∼95–97% top-1 accuracy on a hand-graded eval set and beating a keyword baseline by ∼5 pts.
- Engineered a provider-pluggable LLM/embedding abstraction layer (local stub ↔ Cerebras GPT-OSS 120B / Gemini) behind a single interface, enabling offline development and one-env-var provider swaps with zero call-site changes.
- Implemented anonymous-first authentication (Supabase Auth) with seamless email/password account upgrade, magic-link sign-in, session-ownership authorization, and per-user and per-IP rate limiting via an atomic Postgres RPC.

Weather Forecast App — July 2025 – Aug 2025
React, Node.js, Express.js, Vite, Material UI, REST API

- Built a full-stack weather app (React + Node.js/Express) consuming the OpenWeatherMap API for real-time conditions and 5-day geolocation forecasts, with a RESTful controller-service backend, CORS middleware, and secure dotenv key management.
- Developed custom React hooks and Context API for state management, React Router for client-side routing, and Axios for asynchronous data fetching.